

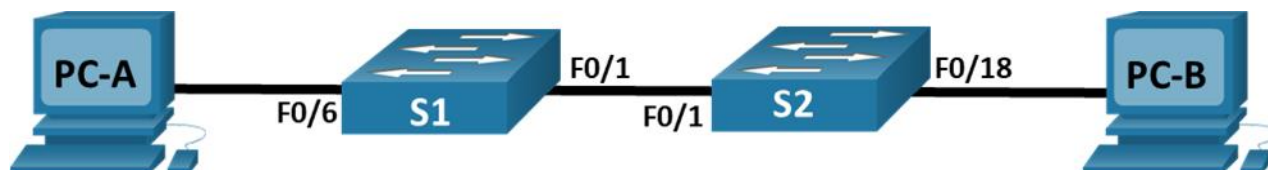
Lab 3.6.2 - Implement VLANs and Trunking



This lab has been updated for use on NETLAB+.

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Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
S1	VLAN 10	192.168.10.11	255.255.255.0
	VLAN 20	192.168.20.11	255.255.255.0
	VLAN 30	192.168.30.11	255.255.255.0
S2	VLAN 10	192.168.10.12	255.255.255.0
PC-A	NIC	192.168.20.13	255.255.255.0
PC-B	NIC	192.168.30.13	255.255.255.0

VLAN Table

VLAN	Name	Interface Assigned
10	Management	S1: VLAN 10 S2: VLAN 10
20	Sales	S1: VLAN 20 and F0/6
30	Operations	S1: VLAN 30 S2: F0/18
999	ParkingLot	S1: F0/2-5, F0/7-24, G0/1-2 S2: F0/2-17, F0/19-24, G0/1-2
1000	Native	N/A

Objectives

Part 1: Configure Basic Device Settings

Part 2: Create VLANs and Assign Switch Ports

Part 3: Configure an 802.1Q Trunk between the Switches

Background / Scenario

Modern switches use virtual local-area networks (VLANs) to improve network performance by separating large Layer 2 broadcast domains into smaller ones. VLANs address scalability, security, and network management. In general, VLANs make it easier to design a network to support the goals of an organization. Communication between VLANs requires a device operating at Layer 3 of the OSI model.

VLAN trunks are used to span VLANs across multiple devices. Trunks allow the traffic from multiple VLANs to travel over a single link, while keeping the VLAN identification and segmentation intact.

In this lab, you will create VLANs on both switches in the topology, assign VLANs to switch access ports, verify that VLANs are working as expected and create VLAN trunks between the two switches.

Note: The switches used with CCNA hands-on labs are Cisco Catalyst 2960s with Cisco IOS Release 15.2(2) (lanbasek9 image). Other switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Instructions

Part 1: Configure Basic Device Settings

In Part 1, you will configure basic settings on the PC hosts and switches.

Step 1: Configure basic settings for each switch.

- Console into the switch and enable privileged EXEC mode.
- Assign a device name to each switch.
- Disable DNS lookup.
- Assign **class** as the privileged EXEC encrypted password.
- Assign **cisco** as the console password and enable login.
- Assign **cisco** as the VTY password and enable login.
- Encrypt the plaintext passwords.
- Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- Copy the running configuration to the startup configuration.

Step 2: Configure PC hosts.

Refer to the Addressing Table for PC host address information.

Part 2: Create VLANs and Assign Switch Ports

In Part 2, you will create VLANs as specified in the table above on both switches. You will then assign the VLANs to the appropriate interface. The **show vlan brief** command is used to verify your configuration settings. Complete the following tasks on each switch.

Step 1: Create VLANs on both switches.

- Create and name the required VLANs on each switch from the table above.
- Configure the management interface on each switch using the IP address information in the Addressing Table.

- c. Assign all unused ports on the switch to the ParkingLot VLAN, configure them for static access mode, and administratively deactivate them.

Step 2: Assign VLANs to the correct switch interfaces.

- a. Assign used ports to the appropriate VLAN (specified in the VLAN table above) and configure them for static access mode.
- b. Verify that the VLANs are assigned to the correct interfaces.

Part 3: Configure an 802.1Q Trunk Between the Switches

In Part 3, you will manually configure interface F0/1 as a trunk.

Step 1: Manually configure trunk interface F0/1.

- a. Change the switchport mode on interface F0/1 to force trunking. Make sure to do this on both switches.
- b. Set the native VLAN to 1000 on both switches.
- c. As another part of trunk configuration, specify that only VLANs 10, 20, 30, and 1000 are allowed to cross the trunk.
- d. Issue the **show interfaces trunk** command to verify trunking ports, the native VLAN and allowed VLANs across the trunk.

Step 2: Verify connectivity.

Verify connectivity within a VLAN. For example, PC-A should be able to ping S1 VLAN 20 successfully.

Were the pings from PC-B to S2 successful? Explain.

Type your answers here.

No, the pings were not successful because S2 does not have VLAN 30 configured (and does not have an IP address), which is what the interface f0/18 is on and PC-B is connected to.

Router and Switch Interface Summary Table

Router / Switch Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1800	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2801	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
2811	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2960	Fast Ethernet 0/1 (F0/1)	Fast Ethernet 0/2 (F0/2)	n/a	n/a

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Router / Switch Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
3560	Fast Ethernet 0/1 (F0/1)	Fast Ethernet 0/2 (F0/2)	n/a	n/a
3650	Gigabit Ethernet 1/0/1 (G1/0/1)	Gigabit Ethernet 1/0/2 (G1/0/2)	n/a	n/a
4221	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
4300	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.